The Periodic Table of the Elements

1A												3A	4A	5A	6A	7A	8A
1 H Hydrogen 1.01	2A																2 He Helium 4.00
3 Li Lithium 6.94	4 Be Beryllium 9.01											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18
11 Na Sodium 22.99	12 Mg Magnesium 24.31	3B	4B	5B	6B	7B	----------8B----------			1B	2B	13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.07	17 Cl Chlorine 35.45	18 Ar Argon 39.95
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.73	32 Ge Germanium 72.61	33 As Arsenic 74.92	34 Se Selenium 78.09	35 Br Bromine 79.90	36 Kr Krypton 84.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.94	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.6	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.33	57 La Lanthanum 138.91	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platnium 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine 209.99	86 Rn Radon 222.02
87 Fr Francium (223)	88 Ra Radium (226)	89 Ac Actinium (227)	104 Rf Rutherfordium (261)	105 Db Dubnium (262)	106 Sg Seaborgium (266)	107 Bh Bohrium (264)	108 Hs Hassium (269)	109 Mt Meitnerium (268)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (280)	112 Cn Copernicium (277)	113 Nh Nihonium (284)	114 Fl Flerovium (289)	115 Mc Moscovium (288)	116 Lv Livermorium (298)	117 Ts Tennessine (294)	118 Og Oganesson (294)

Atomic Number
Element Symbol
Element Name
Average Atomic Mass*

1 H Hydrogen 1.01

*If this number is in parentheses, then it refers to the atomic mass of the most stable isotope.

58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium 144.91	62 Sm Samarium 150.36	63 Eu Europium 151.97	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thullum 168.93	70 Yb Ytterbium 173.04	71 Lu Lutetium 174.97
90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237.05	94 Pu Plutonium 244.06	95 Am Americium 243.06	96 Cm Curium 247.07	97 Bk Berkellum 247.07	98 Cf Californium 251.08	99 Es Einsteinium (254)	100 Fm Fermium 257.10	101 Md Mendelevium 258.1	102 No Nobellum 259.10	103 Lr Lawrencium (262)

States @ room temp
Solid
Liquid
Gas

Polyatomic Ion Chart (and single charge transition metals)

1+		1-		2-		3-	
ammonium	NH_4^+	acetate	$C_2H_3O_2^-$	carbonate	CO_3^{2-}	borate	BO_3^{3-}
silver	Ag^+	hypochlorite	ClO^-	chromate	CrO_4^{2-}	phosphate	PO_4^{3-}
		chlorite	ClO_2^-	dichromate	$Cr_2O_7^{2-}$	phosphite	PO_3^{3-}
2+		chlorate	ClO_3^-	oxalate	$C_2O_4^{2-}$		
cadmium	Cd^{2+}	perchlorate	ClO_4^-	sulfite	SO_3^{2-}		
zinc	Zn^{2+}	cyanide	CN^-	sulfate	SO_4^{2-}		
		hydroxide	OH^-	silicate	SiO_3^{2-}		
1-		iodate	IO_3^-	tartrate	$C_4H_4O_6^{2-}$		
amide	NH_2^-	nitrite	NO_2^-	thiosulfate	$S_2O_3^{2-}$		
cyanate	CNO^-	nitrate	NO_3^-	selenate	SeO_4^{2-}		
bisulfite	HSO_3^-	permanganate	MnO_4^-	arsenate	AsO_4^{2-}		
bicarbonate	HCO_3^-	thiocyanate	SCN^-	persulfate	$S_2O_8^{2-}$		
bromate	BrO_3^-						

Activity Series of Metals (*with Hydrogen*)

More Active → Less Active

Li K Ba Ca Na Mg Al Mn Zn Cr Fe Cd Co Ni Sn Pb H_2 Cu Hg Ag Pt Au

Electronegativity values of the elements (Pauling scale)

H 2.1																	He
Li 1.0	Be 1.5												B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
Na 0.9	Mg 1.2												Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 3.0	
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6	
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Ti 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	Rn 2.4	
Fr 0.7	Ra 0.7	Ac 1.1																

Ce 1.1	Pr 1.1	Nd 1.1	Pm 1.1	Sm 1.1	Eu 1.1	Gd 1.1	Tb 1.1	Dy 1.1	Ho 1.1	Er 1.1	Tm 1.1	Yb 1.1	Lu 1.2
Th 1.3	Pa 1.5	U 1.7	Np 1.3	Pu 1.3	Am 1.3	Cm 1.3	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.3	Lr

SOLUBILITY CHART FOR COMMON IONIC COMPOUNDS IN WATER

	H^+	NH_4^+	Li^+	Na^+	K^+	Mg^{2+}	Ca^{2+}	Ba^{2+}	Al^{3+}	Co^{2+}	$Fe^{2/3+}$	Cu^{2+}	Ag^+	Zn^{2+}	Pb^{2+}	Ni^{2+}
F^-	S	S	S	S	S	INSOL	INSOL	SS	S	S	SS	S	S	S	INSOL	S
Cl^-	S	S	S	S	S	S	S	S	S	S	S	S	INSOL	S	INSOL	S
Br^-	S	S	S	S	S	S	S	S	S	S	S	S	INSOL	S	SS	S
I^-	S	S	S	S	S	S	S	S	S	S	S	INSOL	INSOL	S	INSOL	S
OH^-	H_2O	D	S	S	S	INSOL	SS	SS	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL
SO_3^{2-}	D	S	S	S	S	SS	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL
SO_4^{2-}	S	S	S	S	S	S	SS	INSOL	S	S	S	S	SS	S	INSOL	S
CO_3^{2-}	D	S	S	S	S	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL
NO_3^-	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
PO_4^{3-}	S	S	S	S	S	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL
CrO_4^{2-}	S	S	S	S	S	S	S	INSOL	SS	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL
$C_2H_3O_2^-$	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S
$C_2O_4^{2-}$	S	S	S	S	S	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL	INSOL

INSOL = insoluble in water SS = slightly soluble in water (solid usually) S = Soluble in water D = decomposes in water